Indian Institute of Management Calcutta

Programme: MBA

Name of the Course: Simulations of Economic Decision Making

Core / Elective: Elective Cap (for Elective): NA

Credit: 1.5 credits

Academic Year & Term: 2026-27, Term VI (post-midterm)

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Guest Faculty (GF):

Introduction to Course /Course Description

The primary objective of this course is to make students think like an economist or role-play as an economic agent. Through a variety of computer-aided in-class simulations the course will focus on teaching students how key ideas of making economic choices can be integrated into managerial decision making. Students would have already been exposed to microeconomics, macroeconomics and India and the World Economy core modules and this elective essentially is designed to give them a hands-on feel of what it is like to be an actual agent making the choices which leads to model predictions learnt in theory. The students will learn from conforming results as well as from divergent results and the class discussion on every simulation and the debrief sessions will focus on understanding the relatability of the broad theoretical ideas in terms of practical applications.

The course will 100% rely on running simulations in the classroom and will abstract away from chalk and talk method of teaching. As such the course will take advantage of digital technology enabled pedagogy to deliver the key learning outcomes. More specifically, a set of interactive tools from the ***HBS simulation, MobLab*** will be used for assisted learning in the classroom. The primary idea is to use experiential learning in economics cross-cutting into managerial applications and other allied disciplines.

Course Objective and Key-takeaways from the course

Objectives of the course:

1. Revisit the basic concepts of Economics in a laboratory environment and see practical applications through simulated experiments.
2. Students role-play as economic agents, such as consumer, producer, players in a game, employer, employee, etc and make real decisions within the simulation and observe outcomes and compare them with theory predictions.
3. Make the learning of economics more memorable through experiential learning approach.
4. The course is positioned in a way to emphasize the complementarity of economic decision-making to other fields of management science and to make students appreciate the need to adapt some of the existing models to understand and rationalize behavior in real world transactions.

Key Takeaways from the course:

1. Students will observe how several economic decisions actually operationalize and how their actions can manifest the eventual outcome in a simulated economic transaction.
2. Managerial skills are important in making several of the decisions to ensure that outcomes are favorable for the economic agent, the role of which is being assumed by the student participant of the experiment.
3. In many simulated cases the eventual outcome may diverge from a predicted theoretical equilibrium – as may be the case in real life settings. Students will learn how to adapt to these departures and adjust their managerial decisions to maximize their relevant objective function.

Programme Level Learning Goals

The following Programme Level Learning Goals and objectives shall be attainable through this course:

Learning Goals

GOAL 1: Possess state-of-the-art knowledge of theory and practice in all functional fields of management and the ability to think critically, and apply them to diagnose and find solutions to organisational problems, even in unfamiliar or uncertain situations

- *Objective a*: Attain theoretical and practical understanding of the functions and processes involved in managing an organization
- *Objective b*: Develop critical thinking ability through analytical treatment of important contemporary problems and issues
- *Objective c*: Acquire the ability to design approaches and use relevant models and tools to analyse situations and solve practical business problems
- *Objective d*: Acquire the ability to persuasively recommend managerial solutions issuing from the process of decision analysis applied on managerial situation/s

GOAL 2: Develop and apply ethical values to analyse and solve organisational problems

- *Objective a*: Understand the realm of ethics (morality, values, ethos) and its importance to society
- *Objective c*: Understand the role of manager/leader as an ethical individual and as a professional

GOAL 3: Develop a global perspective that is built upon knowledge of economic, technological, regulatory and political forces that are shaping a new international order in the 21st century

- *Objective b*: Acquire the ability to understand and comprehend issues and practices prevalent in the international arena
- *Objective c*: Develop competence in analysing changes in international and domestic environment
- *Objective d*: Respond to such changes through appropriate organizational decisions

GOAL 4: Develop their capacity to articulate their thinking and ideas through verbal and written communications, at a professional (manager/consultant) level and to deliver the same

- *Objective c*: Acquire the ability to articulate the findings in a coherent and structured format to build a managerial case analysis
- *Objective e*: Acquire the ability to deliver a professional quality formal presentation by exhibiting clarity and appropriate pace, coherent structure, credibility, effective use of visual aids, appropriate body language

GOAL 6: Acquire capacity to apply their professional knowledge and skills to diagnose and resolve business problems in actual organizational settings

- *Objective a*: Acquire the ability to identify organizational problems in both real life settings and through simulated situations
- *Objective b*: Acquire the ability to interpret and apply knowledge of concepts
- *Objective c*: Acquire the ability to design appropriate responses to solve identified problems by using appropriate tools, techniques & framework taught in courses
- *Objective d*: Acquire the ability to apply structured analytical thinking to solve an organizational problem

International Components in your course (if any)

1. All sessions will rely on simulations provided by Harvard Business School Publishing. Many of the simulated experiments will have international contexts.
2. The business implications coming out of the results of most simulations will be discussed in a global context.
3. Additional reading and reference material discussed during debrief sessions of simulation sessions will rely on international examples.

Pre-requisites
Microeconomics, Macroeconomics, India and the World Economy

Required Text Book(s)

No textbook.

Simulation based course. Each session will involve running simulated experiments in the classroom in an attempt to replicate theoretical models and predictions of economics.

Reference Book(s)

No Reference Book.

Course Pack Distribution to students

To be distributed by the MBA office. Additional cases, reading material and/or slides may be distributed to the students from time to time by the instructor or through the program office. Access to the HBS Simulations will be provided through course-pack by the instructor/MBA Office.

Technology enabled learning component for your course

Fully simulation-based course. Access to internet for all sessions in class is required. The simulation runs well on any web browser and also on handheld devices like tablets, iPads etc. However, if students intend to use smartphones – they should check compatibility beforehand and may need to download and install additional apps.

Simulation Required

MobLab Games: All Access, by MobLab. *HBS Product#: ML0020-HTM-ENG*

Session Plan

**** MobLab Simulation will be indicated as MS***

Sl No.	Topic	Intended learning outcome	References
Lecture 1	**Markets** • *Economic Interdependence* • *Supply and Demand* • *Government Policies* • *Welfare Economics and Efficiency*	Students should be able to learn: • About economic interdependence. Students will operate a food truck in the MS and allocate time between preparing burgers and fries to learn about opportunity costs and potential gains from trade. • In the MS on markets, students will randomly be assigned as. Buyers or sellers in a double auction market where sellers ask a price and buyers bid a price to achieve market clearing solutions in the market for oranges or ice creams. Also learn about elasticity through shifts in supply. • In alternative treatments – instructor will simulate binding and non-binding price controls in these markets and introduce taxes to see how buyer and seller participants adjust their "Ask" or their "Bid" to achieve equilibrium outcomes.	1. ***MS: Comparative Advantage Food Truck Game*** (including the PPF game) 2. ***MS: Competitive Market Multi-theme Game*** (double auction market) with following variations: • Shifts in supply • Price Controls • Government Taxes • Deadweight Loss
Lecture 2	**Market Structure - I** • *Monopoly* • *Competition* • *Price Discrimination* • *Entry and Exit*	Students should be able to learn: • About market structures and their impacts. In the monopoly game students play as a monopolist in the pharmaceutical industry and decide how many pallets of pharmaceuticals to ship to the market	3. ***MS: Production Entry and Exit Game*** (including the price takers variant) 4. ***MS: Monopoly Game*** with

		• Students reinforce their earlier learnings on effects of market quantity on equilibrium price and revisit the logic of third-degree price discrimination in a real-world mimicked setting. • In entry and exit game students role play as ride-share drivers and decide whether or not to drive on a given day faced with an upward sloping marginal cost schedule. • Through the entry and exit game students learn about short- and long-run effects on the market and how in a price taking competitive environment firms can survive and sustain their businesses	following variations: • Single market monopoly • Mixed market monopoly • Price discriminating monopoly
Lecture 3	**Market Structure - II** • *Oligopoly* • *Double Marginalization*	Students should be able to learn: • About imperfect competitions and how equilibrium in these markets operationalize. In the Cournot simulation students act as firms and choose quantities of how many scooters to manufacture with a variant which will allow online chats with other playing firms to potentially see if collusion is sustainable and relate to Cournot oligopoly theoretical predictions • In the Bertrand simulation, participants will compete over prices for a homogenous good (class notes for a difficult course) and permeate the Bertrand paradox • In the Stackelberg simulation students will play as either of 2 airline companies, viz, "LeadAir" or "FollowAir" and choose the number of direct flights to learn about the concepts of backward induction and reinforce the idea of the equilbirum in a Stackelberg oligopoly with first mover advantages. • In the double marginalization simulation of the business "CoffeLab Café" student particpants role play as wholeseller and retailer and reinforce their understanding of Stackelberg through an I/O lens.	5. ***MS: Cournot Game*** 6. ***MS: Bertrand Game*** 7. ***MS: Stackelberg Game*** 8. ***MS: Double Marginalization Game***

Lecture 4	**Externalities** • *Positive and Negative Externalities* • *Public Goods* • Common Pool Resources	Students should be able to learn: • About externalities and their role in market failures and potential ways to mitigate these. In the externalities simulation, the students participate as either buyer or seller of Robot Dogs with variants of positive externalities (crime reduction) or potential negative externalities (noise pollution due to excessive barking). The game play makes the students learn how equilibrium outcomes manifest when market failures exist and students also play repeat simulations when corrections are in place such as tax, subsidy or tradeable permits to see how the externality is internalized through their game play. • In the public goods simulations, students are role playing as members of an orchestra deciding how much time to devote to their solo performance vis-à-vis that of the ensemble event to learn about market failures when public good exist. Variations including punishment and failure are used in the sim to make students understand the veracity of market outcomes possible. • Through the Fishery game simulation, students learn about tragedy of commons and become parties to creating such an equilibrium. Variations will include imposition of quotas to mitigate the problem through simulated game play and relate to real world solutions.	9. ***MS: Externalities with Policy Interventions Game*** with following variations: • External Costs • External benefits 10. ***MS: Public Good: Punishment and Rewards Game*** 11. ***MS: Commons: Fishery Game***
Lecture 5	**Managerial Economics and Social Preferences** • *Preliminary Game Theory Applications* • Fairness, Altruism and Reciprocity • Trust in Managerial Contexts	Students should be able to learn: • The concepts of strategic interaction through game theoretic analogs where they decide on push and pull strategies mimicking a standard prisoners' dilemma setting. • Concepts of fairness, altruism and reciprocity through role-playing as agents of a dictator, ultimatum or trust games. • The simulation results will draw upon standard economics predictions or	***12. MS: Push and Pull Game*** ***13. MS: Ultimatum, Dictator and Trust Games***

		departures but through the debrief session students will learn about managerial implications of thinking about non-selfish rational behavior and how fairness, altruism and reciprocal behavior may be relevant in managerial decision making.	
Lecture 6	**Managerial Economics and Asymmetry of Information** • Lessons from Bargaining Models • Principal Agent Problems • Akerlof's Lemon market	Students should be able to learn: • The role of tradeoffs and fairness in continual negotiations environment, relevant for managerial economic decisions. • In the simulation, students will be paired to split a pot of money continually in a bargaining environment and continue offers and counter-offers until the proposal is accepted and through the experiment learn how two-party negotiations are difficult and which tradeoffs one has to live with and compromises, they need to make to crack a deal. • In the multilateral bargaining simulation students play as group of zombies and decide how to split a group of captured humans. This teaches students about the power to propose and the importance of organizing coalitions in decision making. • In the principal-agent simulation, bosses propose a contract to workers under different information scenarios and students learns through their game play about how asymmetric information leads to potential moral hazard and adverse selection problems and managers need to find way to use incentives to minimize these. • Akerlof's Lemon market and the role of market unravelling for used products is learnt through the lemon market buyer simulation.	14. ***MS: Barganing Game*** (including multilateral bargaining game) 15. ***MS: Principal-Agent Game*** 16. ***MS: Lemon market Buyer Game***
Lecture 7	**Economic Decision Making** • Consumer Choice • Risk preferences	Students should be able to learn: • The concepts of marginal utility with fixed budget under a cobb-douglas utility function through simulated experiment where students help the	17. ***MS: Cobb Douglas Game*** 18. ***MS: Bomb Risk Game*** 19. ***MS: Meteor Insurance Game***

	• Insurance and Asymmetric Information	panda allocate money between two types of Sushi. • In the bomb risk game there are 100 boxes in which 1 contains a bomb and students decide how many boxes to open to learn about risk averse and risk loving behavior through simulation results • Adverse selection constrains insurance company and students learn this when they take on the role of an insurance provider firm for meteor-plagued Mars providing meteor insurance. • In the Monty Hall adaptation simulation, students role play as space explorers and attempt to purchase a teleporter to return home and learn the importance of base rate and the Bayes' rule in decision making involving conditional probability.	20. ***MS: Monty Hall Game***
Lecture 8	**Macroeconomics and Business Environment** • Bank Runs • Interest Rates and Inflation • Asset Markets	Students should be able to learn: • The idea of fractional banking and possibility of bank runs through simulated game play where students deposit money in the bank and decide whether to leave the money or withdraw and if too many fellow classmate depositors withdraw, that can lead to bank failure. • About equilibrium interest rates through supply and demand. In the simulated game play, students will decide whether or not to loan their cash to another player from an endowment with a pre-determined return. Essentially, the debrief session will reinforce students' understanding of the Fisher effect. • In simulated game play of bubbles and crashes, students endowed with cash and shared will choose to buy or sell their shares and learn about formation of bubbles and crashes in the economy and debrief will teach them about fundamental and present discounted values of assets.	21. ***MS: Bank Run Game*** 22. ***MS: Bubbles and Crashes Game*** 23. ***MS: Interest Rates and Inflation Game***

Lecture 9	**Public Choice** • Two Candidate Elections • Voter paradox • Voter Turnout	Students should be able to learn: • About location based (spatial) models of economic choice including hoteling problem through a simulation where students will play as candidates in a political race and will decide where to position their campaign on a given policy spectrum. Ideally, debrief will teach students about median voter theorem. • In an alternate simulation, students will play as voters and choose between vote or abstain decisions when voting is costly to learn about predicted turnout and the basic ideas of pivotal voting versus other instrumental motivations of voting. • In a turnout simulation, students decide whether to vote for a soccer club or basketball club at school when only one can be created to learn about underdog voting in democracies like India and its impacts on public policy in the country and voting behavior when elections are expected to be close.	24. ***MS: Two-Candidate Election game*** 25. ***MS: Voter paradox game*** 26. ***MS: Voter Turnout Game***
Lecture 10	**Group Project Presentations**	Students will have to present a synthesis of their learnings and do a small project on an application of Economics based on some experiment and share results.	

Evaluation Components

✓ Components	✓ Weightage
Class Participation	**10%**
Take-Home Assignments	**40%**
Project (Group)	**50%**

Online Course Management (moodle)/course web

Additional material such as slides, references, etc. as and when required will be posted on the course website. The coursepack on HBSP will be created by the instructor and access code for the simulation will be provided. Each session and games from the simulation for the relevant session will be under specific playlists. Students will be provided access to each session-specific playlists.

Mapping for "Simulations of Economic Decision Making" with Programme Level Goals

Goal 1	Goal 2	Goal 3	Goal 4	Goal 5	Goal 6	Goal 7
Critical Thinking a, b, c, d, e	Acquiring Ethical Values a, b ,c ,d, e ,f	Acquiring Global Perspective a, b, c, d,	Ability to Articulate a, b, c, d, e	Technological Competencies a, b, c, d,	Problem Solving a, b, c, d,	Leadership Qualities a, b, c, d,
Simulations of Economic Decision Making (a,b,c,d,e)	Simulations of Economic Decision Making (a, c)	Simulations of Economic Decision Making (b, c, d)	Simulations of Economic Decision Making (c, e)		Simulations of Economic Decision Making (a, b, c, d)	